Shivam Chadha

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Education

BITS Pilani, Goa Campus CGPA: 7.95
MSc. Mathematics and B.E Electronics & Communications *Jul 2019 – Jun 2024*

Experience

QuantumStreet AI Jul'24 - Present
Associate Data Scientist *Bangalore*

- Applied **SHAP** to explain **LGBM/CatBoost** predictions and built a **real-time monitoring system** for **10+ production models**, enabling automated alerts and root-cause analysis
- Designed a **Neo4j knowledge graph** from extracted news concepts and trained **GNN models for edge classification**, advancing graph-based ML solutions
- Migrated **15TB+ total unstructured data** from S3 to OpenSearch. Optimized **daily transfer sub-process**, reducing runtime from **30 minutes to 10 seconds** (180× faster)

Wadhwani AI Jan'24 - Jul'24
ML Intern — Advisor : Dr. Arvind Balachandrasekaran *New Delhi*

- Developed **CatBoost** models to predict **individual student dropout risk** in Gujarat schools using **tabular educational data** (e.g., attendance, grades)
- Developed feature engineering techniques **(DFS, Tree-Based Kernel Embeddings)** for tabular data and evaluated tree vs NN approaches

Amazon Jul'23 - Dec'23
SDE Intern - Ship Tech (Core Trans Tech) *Hyderabad*

- Implemented **throttling** (1000+ TPS) for Geo Carrier Routing Service using **AWS AppConfig**, building a Java caching client for dynamic configuration management
- Secured sensitive customer data by integrating **encryption** for GCRS service logs

Publications

A Machine Learning Framework for Classroom EEG Recording Classification Nov'24
Nanyang Technological University — Advisors - Dr.Yuvaraj Rajamanickam, Dr.Amalin Prince *Remote*

- Developed classification models using **Random Forest, KNN, and MLP** on extracted EEG features, achieving a maximum prediction **accuracy of 78.5%** for classroom teaching methods
- **Published Paper** in Algorithms, 2024, 17(11), 503. DOI: 10.3390/a17110503

A Chaos-Causality Approach to Principled Pruning of Dense Neural Networks Jul'25 (exp)
BITS Pilani, Goa Campus — Advisors: Dr. Snehanshu Saha, Dr. Archana Mathur *Remote*

- Developed novel **pruning** methods (**LEGCNet-FT** and **LEGCNet-PT**) to create pre-set sparse networks that reduce model size while maintaining its performance
- Accepted at *Chaos Theory and Applications* arXiv:2308.09955 [cs.LG]

Projects

FIN-Agent GitHub:Fin-Agent

- Built financial analysis pipeline using **Yahoo Finance API** and **DuckDuckGo news** with **RAG**, generating insights on Indian companies
- Deployed local **Llama3.2** chatbot via **Ollama** with **Gradio** UI for investor Q&A

Skills

Languages: Python , MATLAB, C/C++
Libraries: Numpy, PyTorch, Pandas, Scikit-learn, Matplotlib, Transformers, Hydra
Tools: Git, AWS (S3, AppConfig), Elasticsearch/OpenSearch,, Linux, SSH, Neo4j

Coursework

Data Science: ML, Foundations of Data Science, Applied Statistical Methods, Applied Stochastic Processes
Major: Probability & Statistics, Object Oriented Programming, Linear Algebra, Optimization